MUSIC THEORY

INTERVALS

A Practical Guide for Students of All Ages

Definitions • Explanations • Diagrams • Examples • Practical Exercises

Chapter 1

Introduction to Intervals

Before we can understand chords, scales, or harmony, we must first understand the smallest building block of music: the interval. This chapter introduces what an interval is, how we measure it, and why it matters.

1.1 Definition of an Interval

DEFINITION

An interval is the distance between two musical notes, measured in steps and named with a number (like 2nd, 3rd, or 5th).

EXPLANATION

Think of musical notes as stepping stones on a staircase. Every note has a certain "height" or pitch — some notes sound higher, some sound lower. An interval simply tells us how far apart two of these stepping stones are.

We name intervals using numbers. If you play a note and then count up to another note — including both the note you started on and the note you land on — that count gives you the interval's number. For example, from C to E, we count C (1), D (2), E (3) — so this interval is called a 3rd.

Every interval has two parts: a number (like 3rd or 5th) and a quality (like major, minor, perfect, augmented, or diminished). In this introductory chapter, we will focus on understanding the number — the quality will be explored in a later chapter.

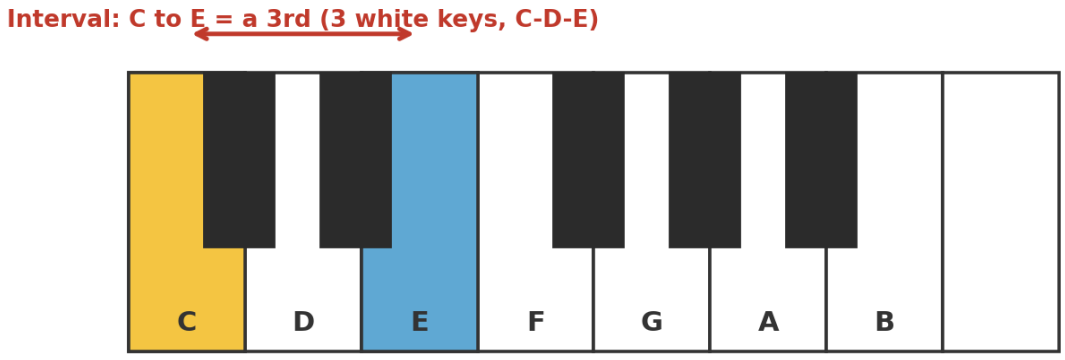


Figure 1.1 — On the keyboard, C to E spans three white keys (C, D, E), so this interval is called a 3rd.

EXAMPLE

Example	Interval Name	How We Count It
C to D	2nd	C (1), D (2)
C to E	3rd	C (1), D (2), E (3)
C to G	5th	C (1), D (2), E (3), F (4), G (5)

Example	Interval Name	How We Count It
C to C (next)	8th (Octave)	C (1) through C (8)

PRACTICAL EXERCISE

1. Using a piano, keyboard app, or the diagram above, find and name the interval from D to G.
2. Name the interval from E to C (count downward this time).
3. Sing or play any two notes on an instrument. Count the letter names to find the interval number.
4. Write down three pairs of notes from your favourite song and label each interval by number.

1.2 Distance Between Two Notes

DEFINITION

The distance between two notes is measured by counting every letter name from the first note to the second note, including both notes themselves.

EXPLANATION

When musicians talk about the "distance" between notes, they do not mean physical distance on an instrument — they mean how many letter-name steps separate the two notes musically.

To measure this distance, always include the starting note as "1" and keep counting each letter name (A, B, C, D, E, F, G) until you reach the second note. It does not matter whether the notes are sharp or flat at this stage — only the letter names are counted for the interval number.

This is a little different from how we count in everyday life. If someone asks "how many steps from 1 to 4?", we might say 3. But in music, the distance from note 1 to note 4 is called a 4th — because we count the starting note itself as the first step.

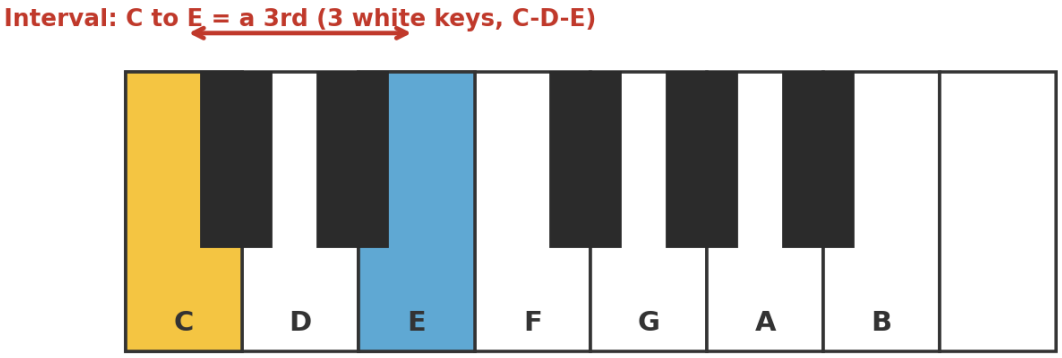


Figure 1.2 — Counting C-D-E on the keyboard gives a distance of 3 letter names, so C to E is a 3rd.

EXAMPLE

Example	Interval Name	How We Count It
F to A	3rd	F (1), G (2), A (3)
G to D	5th	G (1), A (2), B (3), C (4), D (5)
A to B	2nd	A (1), B (2)

PRACTICAL EXERCISE

1. Count the letter-name distance from D to A. What interval number is this?
2. Count the letter-name distance from B to F. What interval number is this?
3. Pick any starting note and count 4 letter names forward. Which note did you land on?
4. Explain in your own words why we count the first note as “1” and not “0”.

1.3 Ascending and Descending Intervals

DEFINITION

An ascending interval moves from a lower note to a higher note. A descending interval moves from a higher note to a lower note.

EXPLANATION

Intervals have a direction. If the second note is higher in pitch than the first note, the interval is ascending — imagine walking up a staircase. If the second note is lower in pitch than the first note, the interval is descending — imagine walking back down.

The interval number (2nd, 3rd, 5th, and so on) stays the same whether you move up or down between the same two letter names. For example, C up to G is a 5th, and G down to C is also a 5th — only the direction is different.

Recognising direction is important for reading melodies. A melody that rises often feels exciting, hopeful, or building in energy, while a melody that falls often feels calming, resolving, or settling down.

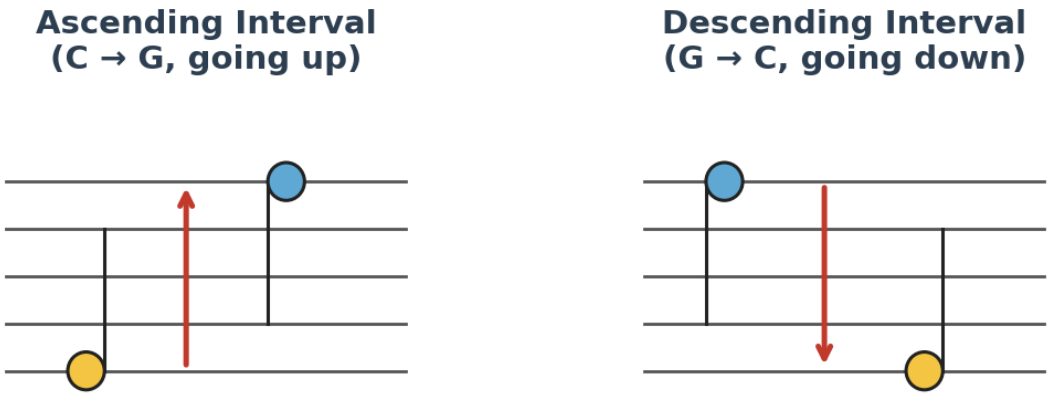


Figure 1.3 — The same two notes, C and G: on the left the melody moves up (ascending), on the right it moves down (descending).

EXAMPLE

Example	Interval Name	How We Count It
C → G	Ascending 5th	Lower note played first, then a higher note
G → C	Descending 5th	Higher note played first, then a lower note
E → C	Descending 3rd	Moving from E down to C

PRACTICAL EXERCISE

1. Play or sing C then G. Is this interval ascending or descending?
2. Play or sing A then D (the D below A). Is this interval ascending or descending?
3. Listen to the first line of any song you know. Does the melody mostly ascend, descend, or do both?
4. Write two short 3-note melodies of your own: one that only ascends, and one that only descends.

1.4 Melodic and Harmonic Intervals

DEFINITION

A melodic interval is formed when two notes are played one after another. A harmonic interval is formed when two notes are played at the same time.

EXPLANATION

So far, we have been looking at notes played one after another — this is called a melodic interval, because melodies are made of notes played in sequence, one at a time.

But two notes can also be played together, at exactly the same moment. When this happens, we call it a harmonic interval, because playing notes together is the foundation of harmony — the study of how notes combine to create chords and rich, layered sound.

Both types of intervals use the same counting rules from Sections 1.1 and 1.2 — the only difference is timing: melodic intervals happen across time, while harmonic intervals happen at a single moment in time.

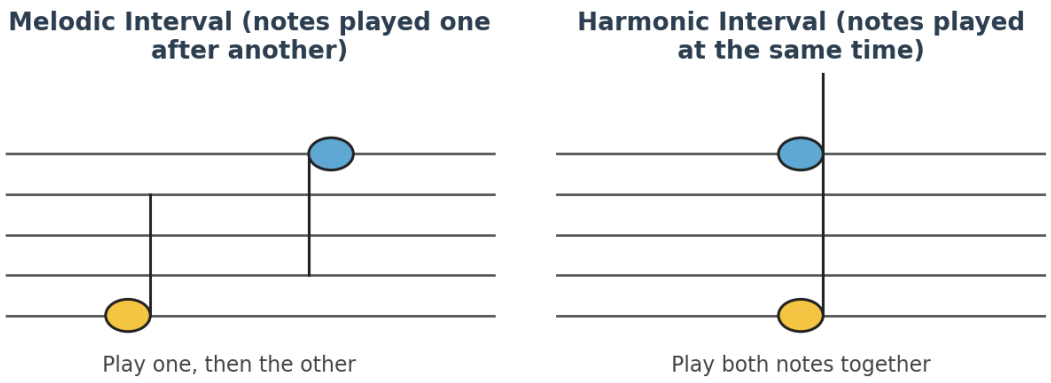


Figure 1.4 — On the left, the two notes are played one after another (melodic). On the right, the same two notes are played together (harmonic).

EXAMPLE

Example	Interval Name	How We Count It
Singing a scale note by note	Melodic interval	Notes happen one after the other
Playing two piano keys together with one hand	Harmonic interval	Notes happen at the same time
A guitarist strumming two notes of a chord together	Harmonic interval	Simultaneous sound

PRACTICAL EXERCISE

1. On any instrument, play C and E one after another. What type of interval is this?
2. On any instrument, play C and E together at the same time. What type of interval is this now?
3. Think of a guitar chord you know. Are the notes in a chord melodic or harmonic intervals?
4. Hum a short 2-note melody, then try to hum both notes as one blended sound. Notice the difference in how each one feels.

1.5 Why Intervals Are Important

DEFINITION

Intervals are the basic building blocks of melody, harmony, chords, and scales — nearly everything in music is built from them.

EXPLANATION

Once you understand intervals, you begin to understand how music itself is built. A scale is really just a specific pattern of intervals played one after another. A chord is really just a specific pattern of intervals played together.

Musicians use interval names to communicate quickly and clearly. Instead of saying “play the note that is two letter names higher”, a musician can simply say “play a 3rd higher”, and every trained musician will understand exactly what to do.

Training your ear to recognise intervals also helps you play music by ear, sing in tune, write your own melodies, and understand why certain combinations of notes sound happy, sad, tense, or peaceful. This is why intervals are often taught as the very first topic in music theory — everything else is built on top of this foundation.

EXAMPLE

Example	Interval Name	How We Count It
Major scale	Built from a fixed pattern of 2nds	Whole and half step intervals in sequence
Major chord	Built from stacked 3rds	Root + major 3rd + minor 3rd
Octave	The interval that defines “same note, higher or lower”	8 letter names apart

PRACTICAL EXERCISE

1. List three places in music (songs, scales, chords) where you think intervals are being used, even if you haven't noticed it before.
2. Ask a friend to play two notes. Try to guess the interval number just by listening.
3. In your own words, explain to someone else why learning intervals is useful for a musician.
4. Look back at Sections 1.1 to 1.4. Write one sentence summarising what you learned from each.

Chapter 2

Understanding Note Distance

In Chapter 1, we learned that an interval is the distance between two notes. Now we go one level deeper and learn the two smallest units musicians use to measure that distance: the semitone and the whole tone. We will also learn two ways notes can move — chromatically and diatonically — and see all of this laid out clearly on the keyboard.

2.1 Semitone

DEFINITION

A semitone (also called a half step) is the smallest distance between two notes in Western music.

EXPLANATION

On a piano keyboard, a semitone is the distance from any key to the very next key beside it — whether that next key is black or white. There is no key in between them.

Most of the time, moving a semitone means moving from a white key to its neighbouring black key (like C to C#). But in two special places on the keyboard — E to F, and B to C — there is no black key in between at all, so two white keys sit right next to each other and are still only a semitone apart.

The semitone is important because it is the basic unit of measurement in music. Every other interval, including the whole tone, can be described as a certain number of semitones.

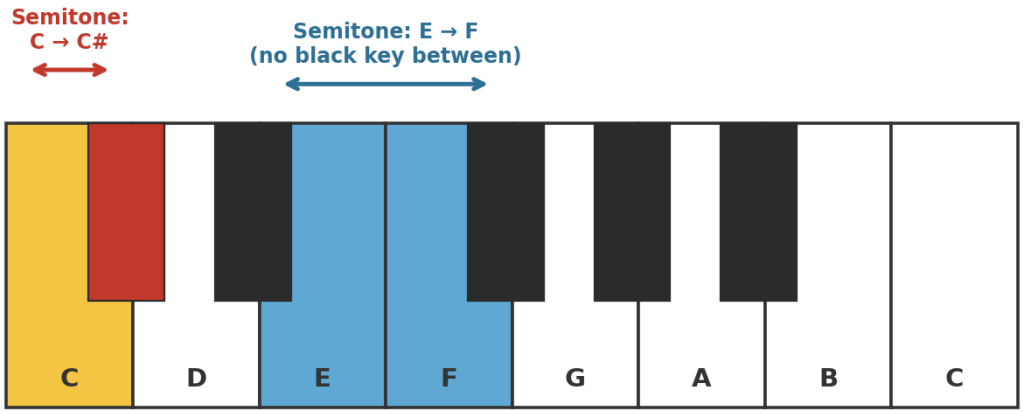


Figure 2.1 — Two examples of a semitone: C to C# (with a black key in between) and E to F (no black key in between).

EXAMPLE

Example	Interval Name	How We Count It
C to C#	Semitone	Adjacent keys, black key involved
E to F	Semitone	Adjacent white keys, no black key between
B to C	Semitone	Adjacent white keys, no black key between

PRACTICAL EXERCISE

1. On a keyboard or the diagram above, find and play all the semitone pairs that use a black key between them within one octave.
2. Name the two places on the keyboard where a semitone happens between two white keys with no black key between.
3. Play G and G#. Is this pair a semitone? Explain why.
4. Why do you think the semitone is called the “smallest” distance in music?

2.2 Whole Tone

DEFINITION

A whole tone (also called a whole step) is a distance equal to two semitones.

EXPLANATION

If a semitone is one small step, a whole tone is simply two of those steps put together. To move a whole tone from any note, skip over one key (black or white) and land on the key after it.

For example, starting on C: one semitone up is C#, and one more semitone up from C# is D. So C to D is a whole tone — it “passes through” C# along the way, even though C# is not played.

Whole tones are extremely common in music. In fact, most of the steps in a major scale are whole tones, with only two semitones mixed in — we will see this clearly in Section 2.4 on diatonic movement.

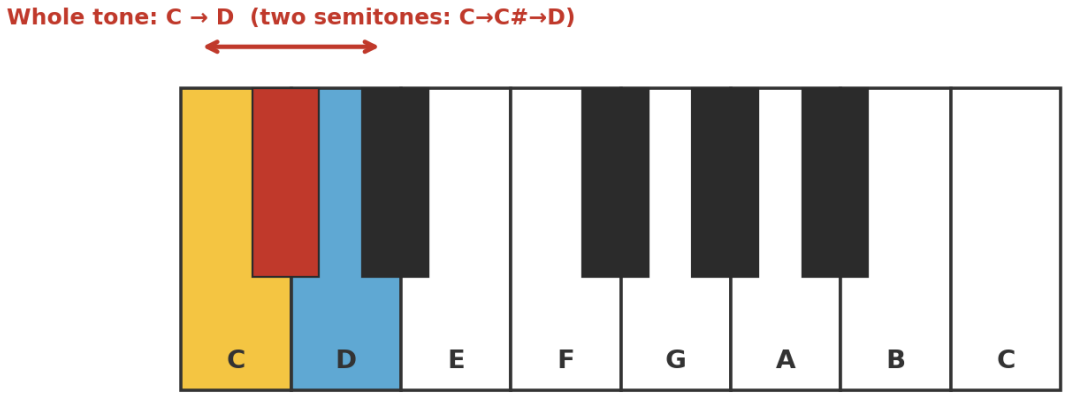


Figure 2.2 — C to D is a whole tone because it equals two semitones (C to C#, then C# to D).

EXAMPLE

Example	Interval Name	How We Count It
C to D	Whole tone	C → C# → D (2 semitones)
F to G	Whole tone	F → F# → G (2 semitones)

Example	Interval Name	How We Count It
A to B	Whole tone	A → A# → B (2 semitones)

PRACTICAL EXERCISE

1. Starting on D, move up one whole tone. Which note do you land on?
2. Starting on G, move up one whole tone. Which note do you land on?
3. Is the distance from E to F a whole tone or a semitone? Explain your answer using what you learned in Section 2.1.
4. Play two whole tones in a row starting on C (C to D, then D to E). What do you notice about the sound?

2.3 Chromatic Movement

DEFINITION

Chromatic movement means moving from one note to the very next note, semitone by semitone, using every available key — black and white — without skipping any.

EXPLANATION

The word “chromatic” comes from a Greek word meaning “colour” — musicians use it to describe a scale or melody that uses all twelve notes available within an octave, including every black key, like adding in every possible colour.

When a melody or scale moves chromatically, it climbs (or descends) in semitones with no gaps at all. This creates a smooth, sliding sound that is very different from the more familiar sound of a major or minor scale.

Chromatic movement is often used to create tension, drama, or a sense of sliding between two points in music — you can hear it in suspenseful film music, jazz improvisation, and blues.

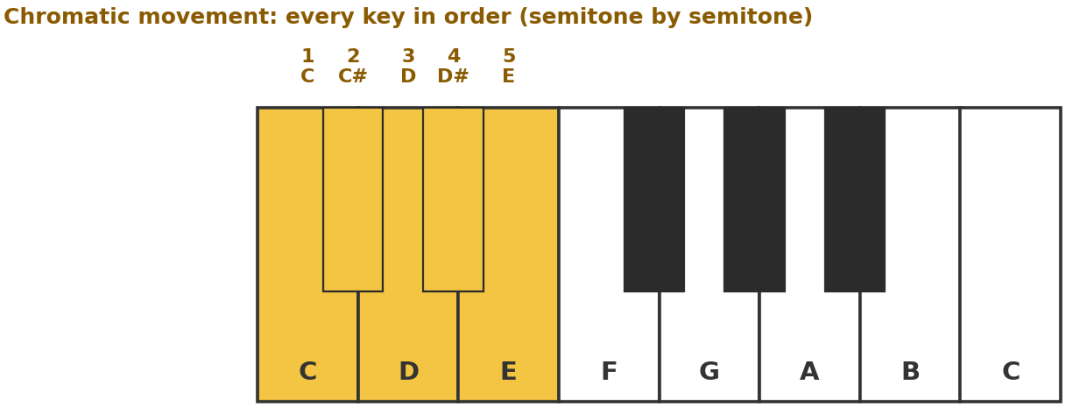


Figure 2.3 — Chromatic movement from C to E uses every key in between: C, C#, D, D#, E — five notes, four semitones.

EXAMPLE

Example	Interval Name	How We Count It
C, C#, D, D#, E	Chromatic movement	Every key used, no notes skipped
G, F#, F, E	Chromatic movement (descending)	Moving down semitone by semitone

PRACTICAL EXERCISE

1. Starting on C, play every key going up chromatically until you reach the next C (the octave). How many notes did you play in total, including the first and last?
2. Try playing a chromatic run from F to A. Write out every note you played in order.
3. Listen to a piece of suspenseful movie music. Can you hear any chromatic (sliding) movement?
4. Explain in your own words why chromatic movement might sound different from a normal major scale.

2.4 Diatonic Movement

DEFINITION

Diatonic movement means moving only through the notes that belong to a particular scale, skipping over any notes that do not belong to it.

EXPLANATION

While chromatic movement uses every key without exception, diatonic movement is more selective — it only uses the notes that belong to a specific scale, such as the C major scale (C, D, E, F, G, A, B).

The C major scale is a good example because it uses only the white keys. If you play all the white keys from C to the next C in order, you are moving diatonically through the C major scale.

Notice that the distances between these notes are not all equal — most steps are whole tones, but two steps (E to F, and B to C) are semitones. This specific pattern of whole tones and semitones — Whole, Whole, Half, Whole, Whole, Whole, Half — is what gives the major scale its familiar, pleasant sound.

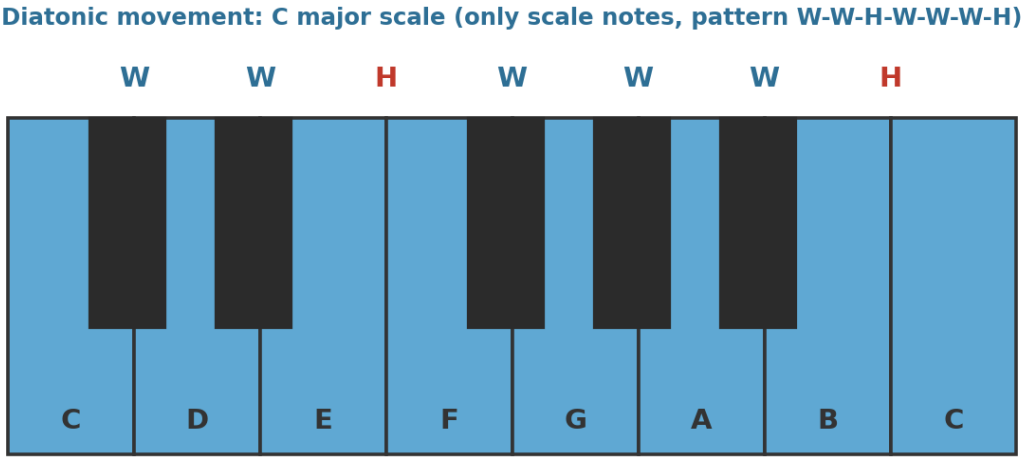


Figure 2.4 — The C major scale moves diatonically through only the white keys, following the pattern Whole-Whole-Half-Whole-Whole-Whole-Half.

EXAMPLE

Example	Interval Name	How We Count It
C to D	Diatonic step (Whole tone)	Both notes belong to C major
E to F	Diatonic step (Semitone)	Both notes belong to C major
C to C#	NOT diatonic in C major	C# does not belong to the C major scale

PRACTICAL EXERCISE

1. Write out the C major scale from memory: C, D, E, __, __, __, __, C.
2. Identify which two steps in the C major scale are semitones (half steps) rather than whole tones.
3. Is the note F# part of the C major scale? Is a move from F to F# diatonic or chromatic?
4. Compare chromatic movement (Section 2.3) and diatonic movement (this section) in your own words. What is the main difference?

2.5 Intervals on Keyboard

DEFINITION

Every interval can be measured on the keyboard by counting the number of semitones between the two notes.

EXPLANATION

Now that we understand semitones and whole tones, we can measure any interval precisely — not just by counting letter names (as we did in Chapter 1), but by counting the exact number of semitones between two notes.

This semitone count is what gives an interval its full identity. For example, a 3rd can be 3 semitones (called a minor 3rd) or 4 semitones (called a major 3rd) — the letter-name count is the same, but the semitone count tells us exactly how the interval sounds.

The keyboard is the best tool for learning this, because black and white keys make every semitone visible. As you look at the diagram below, notice how the number of semitones increases steadily as we move from the root note up to the octave.

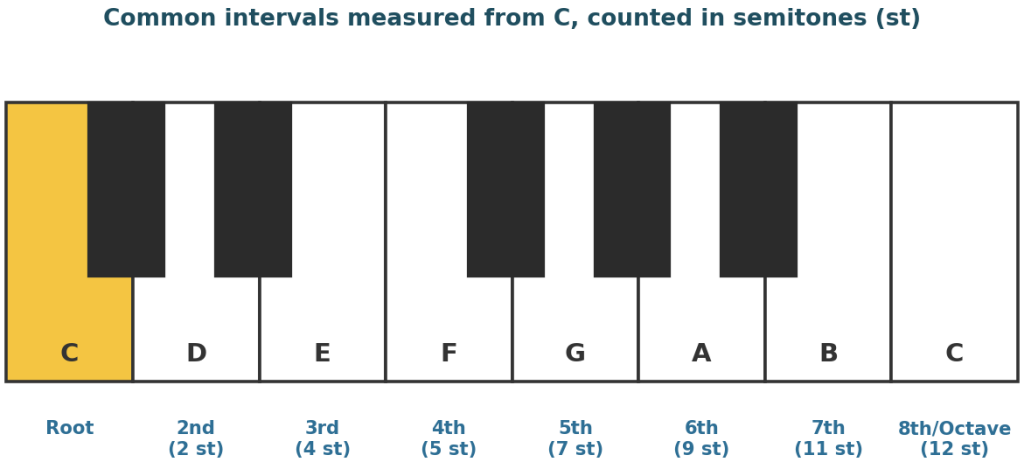


Figure 2.5 — Common intervals measured from C, with their exact semitone counts shown beneath each key.

EXAMPLE

Example	Interval Name	How We Count It
C to E	Major 3rd	4 semitones
C to G	Perfect 5th	7 semitones
C to C (next)	Octave	12 semitones

PRACTICAL EXERCISE

1. Using the diagram, count how many semitones make up the interval from C to A.
2. Using the diagram, count how many semitones make up the interval from C to F.
3. Pick any starting note on a keyboard and count 7 semitones upward. Which interval have you played?
4. Explain why knowing the semitone count of an interval is more precise than only counting letter names.

Chapter 3

Interval Numbers

Every interval we studied in Chapters 1 and 2 has a number — 2nd, 3rd, 5th, and so on. In this chapter, we look closely at each interval number from the smallest (Unison) to the largest within one octave, learn what each one sounds and feels like, and finish with a clear step-by-step method for counting any interval number yourself.

3.1 Unison

DEFINITION

A unison is the interval between two identical notes — the same pitch played or sung twice, with no distance between them at all.

EXPLANATION

A unison is technically the starting point of all intervals — it is what happens before any distance has been added. If two instruments or singers play or sing the exact same note at the exact same time, they are said to be "in unison."

Because there is no difference in pitch, a unison is sometimes called a "zero interval" in terms of distance, but musicians still count it as "1" using the same counting method as every other interval — you will see why in Section 3.9.

Unisons are used often in group singing and orchestras, where multiple performers play or sing the exact same note together to make the sound fuller and stronger, without changing the pitch itself.

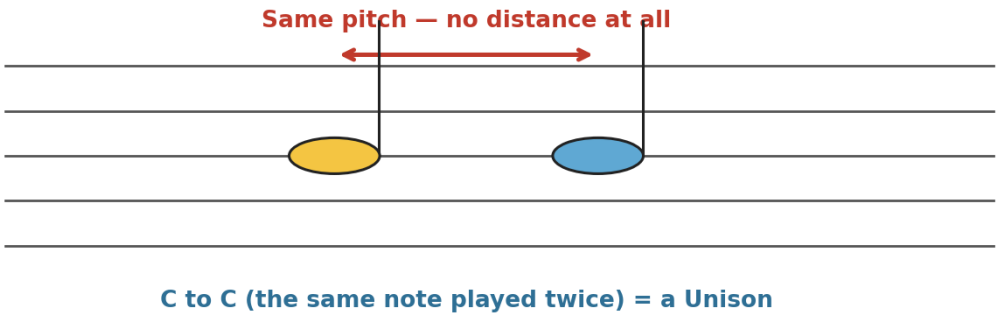


Figure 3.1 — A unison: the same note, C, played twice with no change in pitch.

EXAMPLE

Example	Interval Name	How We Count It
Two singers both singing middle C	Unison	Same pitch, no distance
A violin section playing the same note together	Unison	Reinforces the sound, no interval distance

PRACTICAL EXERCISE

1. Play any note on a keyboard twice in a row. Has the pitch changed at all?
2. Ask a friend to hum the same note as you at the same time. What do you notice about how the combined sound feels?
3. Why do you think orchestras use unison playing to make a sound feel more powerful?
4. Is a unison the smallest interval, or is a semitone smaller? Explain your reasoning.

3.2 2nd

DEFINITION

A 2nd is the interval between a note and the very next letter name above or below it.

EXPLANATION

A 2nd is the smallest interval where two different letter names are involved. It can be a minor 2nd (1 semitone, a very close, tense-sounding distance) or a major 2nd (2 semitones, also called a whole tone, which sounds smoother).

You have already met the 2nd without realising it — it is the interval between each neighbouring note of a scale, like C to D or D to E in the C major scale.

Because a 2nd is such a small step, melodies that move mostly in 2nds tend to sound smooth and connected, almost like walking calmly rather than jumping.

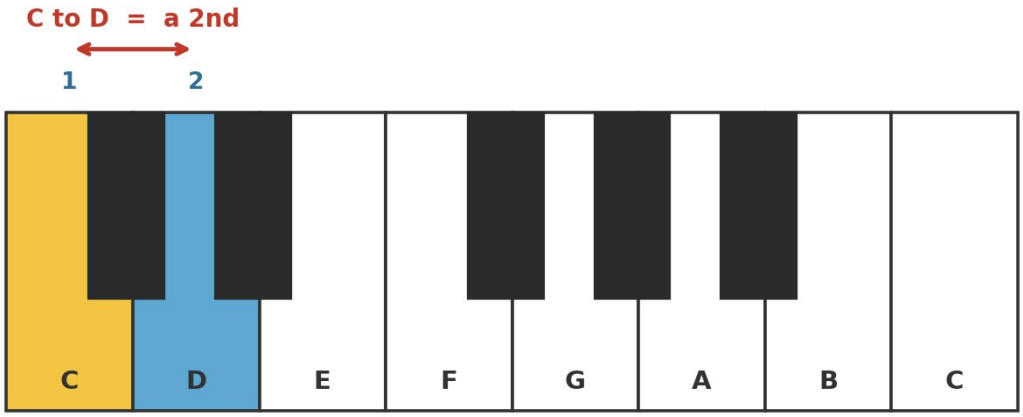


Figure 3.2 — C to D is a 2nd — the interval between one letter name and the very next one.

EXAMPLE

Example	Interval Name	How We Count It
C to D	Major 2nd	2 semitones (whole tone)
E to F	Minor 2nd	1 semitone (semitone)

Example	Interval Name	How We Count It
Twinkle Twinkle Little Star opening notes	Uses 2nds	Smooth, step-by-step movement

PRACTICAL EXERCISE

1. Play C then D. Count the letter names to confirm this is a 2nd.
2. Play E then F. Is this also a 2nd? How many semitones does it contain?
3. Hum the start of a nursery rhyme you know. Can you spot any 2nds (notes right next to each other)?
4. In your own words, describe what a melody built only from 2nds might sound like.

3.3 3rd

DEFINITION

A 3rd is the interval spanning three letter names, counting both the starting and ending note.

EXPLANATION

The 3rd is one of the most important intervals in all of music, because it is the interval that defines whether a chord sounds major (happy, bright) or minor (sad, dark).

A major 3rd spans 4 semitones (like C to E) and has a bright, open sound. A minor 3rd spans 3 semitones (like C to Eb) and has a softer, more emotional sound.

Because of this, whenever you build a simple chord by stacking 3rds on top of a root note, the type of 3rd you choose has a huge effect on the mood of that chord — we will explore this fully in a later chapter on chords.

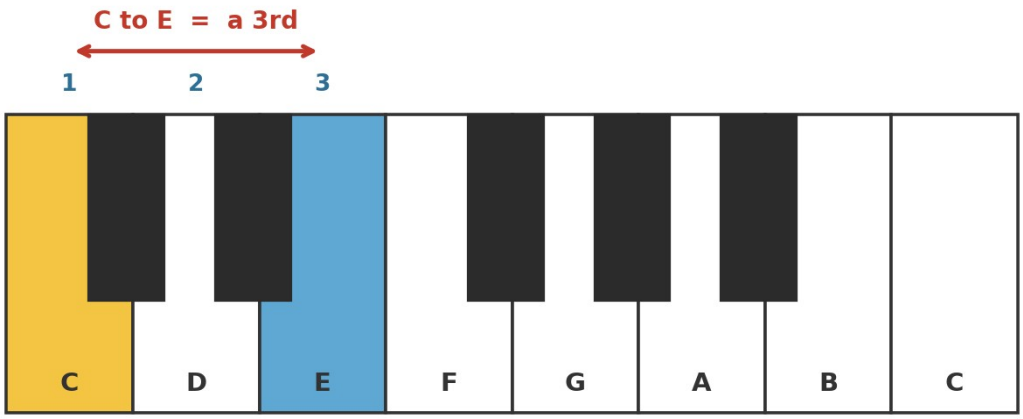


Figure 3.3 — C to E is a 3rd — spanning three letter names: C, D, E.

EXAMPLE

Example	Interval Name	How We Count It
C to E	Major 3rd	4 semitones
C to Eb	Minor 3rd	3 semitones
A major chord (C-E-G)	Built using two stacked 3rds	C to E is a major 3rd, E to G is a minor 3rd

PRACTICAL EXERCISE

1. Play C and E together. Does this sound bright and happy, or sad?
2. Play C and Eb together. How is this different from C and E?
3. Count the letter names from F to A. Confirm that this is also a 3rd.
4. Why do you think the 3rd is called the interval that decides if a chord is major or minor?

3.4 4th

DEFINITION

A 4th is the interval spanning four letter names, counting both the starting and ending note.

EXPLANATION

The 4th has a strong, stable, slightly "hollow" sound compared to the richer sound of a 3rd. Most 4ths you will meet are called perfect 4ths, spanning 5 semitones.

You have almost certainly heard a 4th before — the opening two notes of the wedding tune "Here Comes the Bride" leap upward by a perfect 4th, and so does the well-known melody "Amazing Grace."

The 4th is also very important on guitar: when you tune a standard guitar, most of the strings next to each other are tuned a perfect 4th apart.

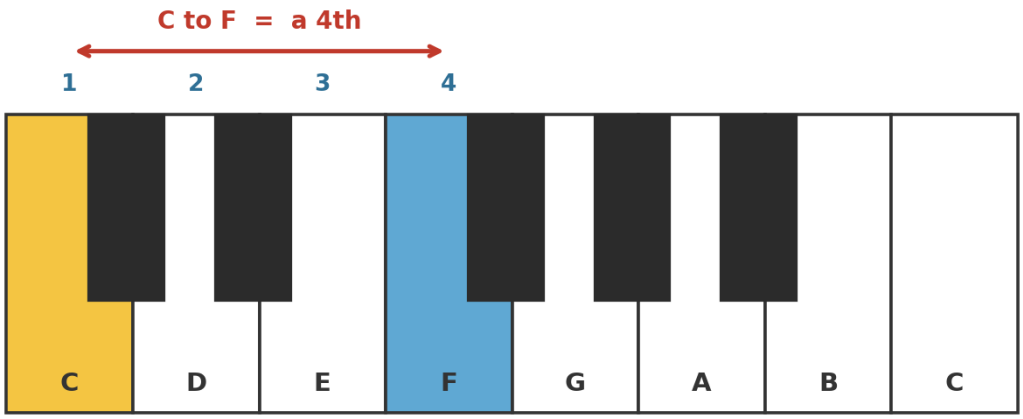


Figure 3.4 — C to F is a 4th — spanning four letter names: C, D, E, F.

EXAMPLE

Example	Interval Name	How We Count It
C to F	Perfect 4th	5 semitones
G to C	Perfect 4th	5 semitones
“Here Comes the Bride” opening leap	Perfect 4th	Recognisable rising leap

PRACTICAL EXERCISE

1. Play C then F. Count the letter names to confirm this is a 4th.
2. Hum the opening two notes of “Here Comes the Bride.” Can you feel the upward leap?
3. If you play guitar or know someone who does, ask them about how the strings are tuned. Do you notice the 4th relationship?
4. Compare the sound of a 3rd (Section 3.3) and a 4th. Which feels more “hollow” or open to you?

3.5 5th

DEFINITION

A 5th is the interval spanning five letter names, counting both the starting and ending note.

EXPLANATION

The perfect 5th (7 semitones) is one of the most stable and pleasant-sounding intervals in music. It sounds strong and “open” without feeling tense.

The 5th is the foundation of the “power chord” used constantly in rock and metal guitar music — a power chord is simply a root note and a 5th played together, sometimes with the octave added too.

The 5th is also the interval used to build the circle of fifths, an important tool in music theory that organises all twelve keys in a circular pattern — you will learn more about this in a later chapter.

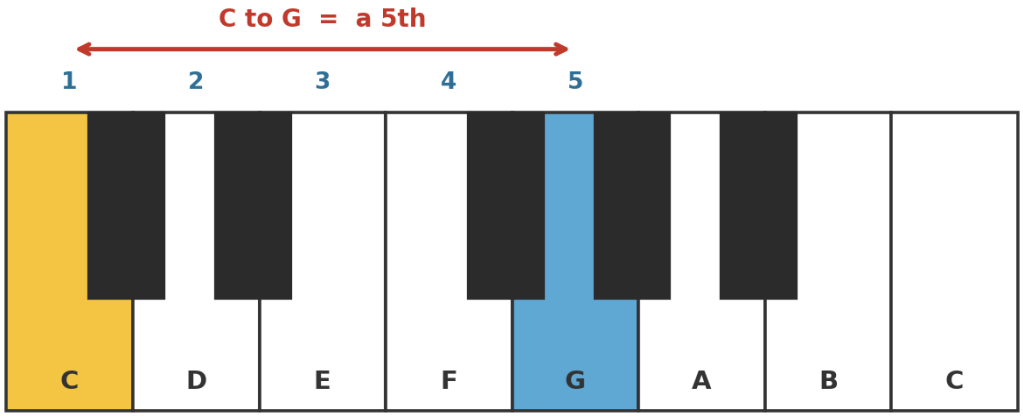


Figure 3.5 — C to G is a 5th — spanning five letter names: C, D, E, F, G.

EXAMPLE

Example	Interval Name	How We Count It
C to G	Perfect 5th	7 semitones
D to A	Perfect 5th	7 semitones
Rock power chord (e.g. C5)	Built from a root and a 5th	Root + perfect 5th, sometimes + octave

PRACTICAL EXERCISE

1. Play C and G together. How would you describe the sound — tense, or open and stable?
2. If you play guitar, try forming a simple power chord (root + 5th). What style of music does it remind you of?
3. Count the letter names from D to A. Confirm this is also a 5th.
4. Why do you think the 5th is considered one of the most “stable” sounding intervals?

3.6 6th

DEFINITION

A 6th is the interval spanning six letter names, counting both the starting and ending note.

EXPLANATION

A 6th is a fairly wide interval with a warm, pleasant sound. Like the 3rd, it comes in two common types: a major 6th (9 semitones) which sounds bright, and a minor 6th (8 semitones) which sounds a little more emotional or wistful.

You may recognise the major 6th from the opening of the folk song “My Bonnie Lies Over the Ocean”, where the melody leaps up by a 6th right at the start.

6ths are commonly used in harmony singing, where a second singer harmonises a melody by singing a 6th above or below the main tune, creating a sweet, pleasing blend.

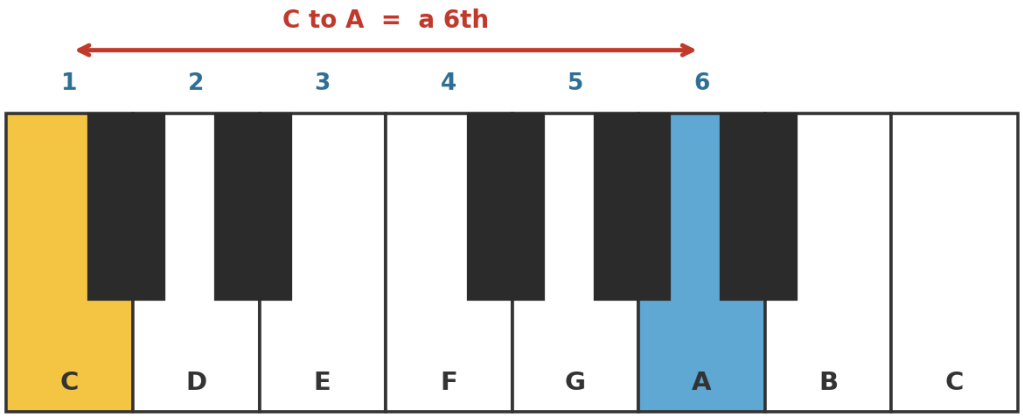


Figure 3.6 — C to A is a 6th — spanning six letter names: C, D, E, F, G, A.

EXAMPLE

Example	Interval Name	How We Count It
C to A	Major 6th	9 semitones
C to Ab	Minor 6th	8 semitones
“My Bonnie Lies Over the Ocean” opening leap	Major 6th	Recognisable wide, warm leap

PRACTICAL EXERCISE

1. Play C then A. Count the letter names to confirm this is a 6th.
2. Compare a major 6th (C to A) with a minor 6th (C to Ab). What difference do you hear?
3. Try singing a melody you know, then have a friend try singing a 6th above or below you at the same time.
4. Why might singers enjoy using 6ths for harmony rather than very small intervals like 2nds?

3.7 7th

DEFINITION

A 7th is the interval spanning seven letter names, counting both the starting and ending note — the widest interval before reaching the octave.

EXPLANATION

A 7th is a fairly tense-sounding interval, especially the major 7th (11 semitones), which sits just one semitone below the octave and creates a strong pull, or “leading” feeling, toward the octave note above it.

The minor 7th (10 semitones) is extremely common in jazz, blues, and pop music, and gives chords a relaxed, slightly unresolved, “bluesy” character.

Because 7ths sound tense or unresolved on their own, they are often used to create a feeling of motion in music — a sense that the music wants to continue on to another note or chord, usually the octave or the root.

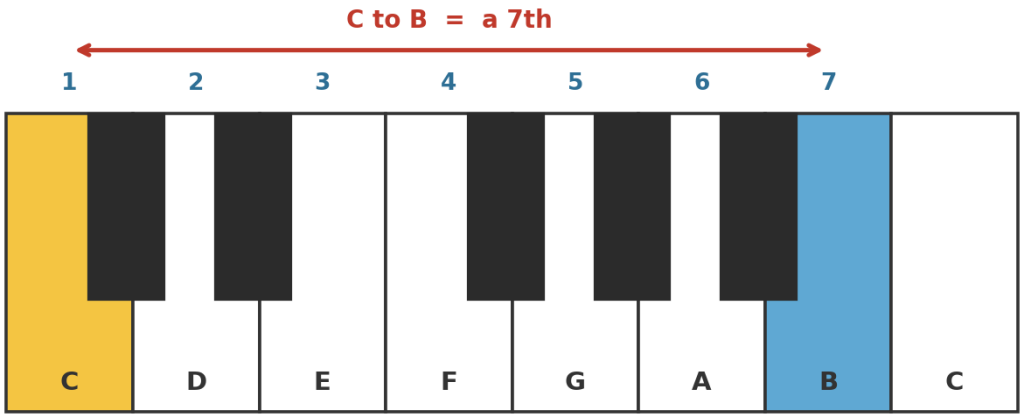


Figure 3.7 — C to B is a 7th — spanning seven letter names: C, D, E, F, G, A, B.

EXAMPLE

Example	Interval Name	How We Count It
C to B	Major 7th	11 semitones
C to Bb	Minor 7th	10 semitones
Jazz “dominant 7th” chords	Built using a minor 7th	Common in blues and jazz harmony

PRACTICAL EXERCISE

1. Play C then B. Count the letter names to confirm this is a 7th.
2. Play C and B together. Does it feel resolved and calm, or tense and unresolved?
3. Compare a major 7th (C to B) with a minor 7th (C to Bb). Which sounds more “bluesy” to you?
4. Why do you think musicians describe the 7th as an interval that creates a “pull” toward another note?

3.8 Octave

DEFINITION

An octave is the interval spanning eight letter names, from a note to the next note of the same letter name — the point where the pattern of notes repeats.

EXPLANATION

The octave (12 semitones) is one of the most special intervals in music. Two notes an octave apart share the same letter name and sound remarkably alike — almost like the “same” note, just higher or lower — even though their pitches are clearly different.

This special relationship exists because of how sound works: a note one octave higher vibrates at exactly double the frequency of the note below it. This is a physical, scientific fact, not just a musical convention.

The octave is also why musical notes only use seven letter names (A to G) before repeating — once you reach the octave, the entire pattern of letter names simply starts over again at a higher or lower pitch.

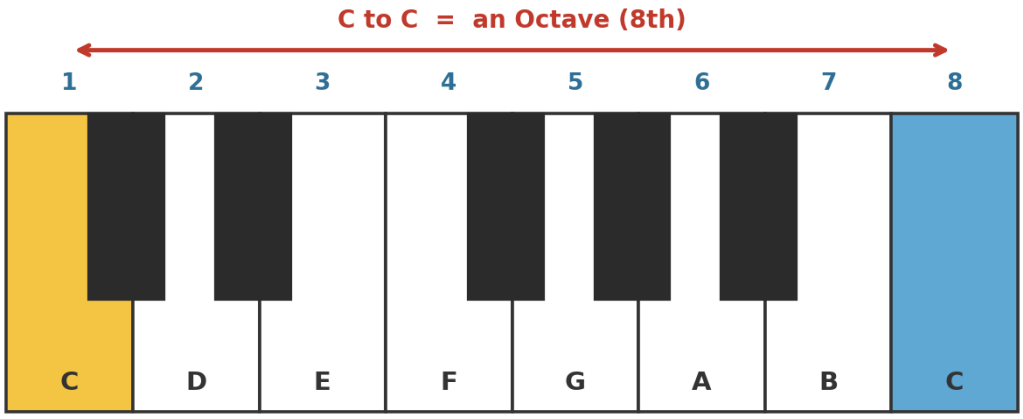


Figure 3.8 — C to the next C is an octave — spanning eight letter names and returning to the same letter name.

EXAMPLE

Example	Interval Name	How We Count It
C to C (next)	Octave	12 semitones, double the frequency
A man and a woman singing “the same tune” comfortably	Often an octave apart	Same melody, different pitch range
Piano keyboard pattern	Repeats every octave	Same white/black key pattern repeats

PRACTICAL EXERCISE

1. Play C, then play the next C above it. Do the two notes sound similar, even though one is higher?
2. Look at a full piano keyboard (in person or a picture). Can you see the same pattern of black keys repeating?
3. Why do you think musicians only use seven letter names (A to G) instead of a new name for every key?
4. Explain in your own words why an octave feels like “the same note, just higher or lower.”

3.9 How to Count Interval Numbers

DEFINITION

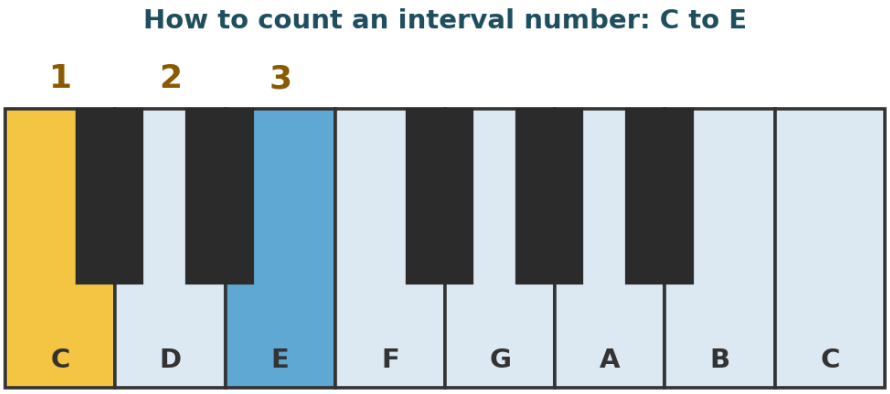
To find the number of any interval, count every letter name from the first note to the second note, including both notes, and the final count is the interval number.

EXPLANATION

Now that you have met every interval number from Unison to Octave, here is the simple, repeatable method for working out the number of any interval, every time.

Always start counting from the first note, and always call that first note “1” — never “0.” Then count upward (or downward) through every letter name in order until you reach the second note. Do not skip any letter names, even if a sharp or flat is involved.

The last number you land on is your answer. It does not matter whether the notes involve sharps or flats — the interval number only cares about letter names. The sharp or flat only affects the interval's quality (major, minor, and so on), which is a topic for a later chapter.



Step 1: Start counting from the FIRST note. Call it “1” (not “0”).

Step 2: Count every letter name upward: C=1, D=2, E=3 — include both notes.

Step 3: The last number you reach is the interval number. Here, that number is 3, so C to E is a 3rd.

Figure 3.9 — The three-step method for counting an interval number, shown using C to E.

EXAMPLE

Example	Interval Name	How We Count It
D to G	4th	D (1), E (2), F (3), G (4)
E to C (next)	6th	E (1), F (2), G (3), A (4), B (5), C (6)
Bb to Eb	4th	B (1), C (2), D (3), E (4) — flats do not change the count

PRACTICAL EXERCISE

1. Use the three-step method to find the interval number from F to B.
2. Use the three-step method to find the interval number from A to F.
3. A friend tells you F# to B is a 4th. Using only letter names, check if they are correct.
4. Summarise the three-step counting method in your own words, as if you were teaching it

to a younger student.

Chapter 4

Interval Qualities

In Chapter 3, we learned how to find an interval's number (like 3rd or 5th) by counting letter names. But the number alone does not tell us the exact size of an interval in semitones — for that, we need a quality. In this chapter, we learn the five interval qualities — Major, Minor, Perfect, Augmented, and Diminished — which, combined with the number, give every interval its full and precise name.

4.1 Major

DEFINITION

A major interval is the larger of the two common sizes for 2nds, 3rds, 6ths, and 7ths, and is the size naturally found above the tonic (starting note) of a major scale.

EXPLANATION

The word “major” means “larger”, and it is used to describe 2nds, 3rds, 6ths, and 7ths that are one semitone bigger than their minor equivalent (which we will meet in the next section).

The easiest way to find every major interval is to build a major scale from your starting note and see which notes appear in it. For example, in the C major scale (C, D, E, F, G, A, B, C), the interval from C up to any other scale note that is a 2nd, 3rd, 6th, or 7th is automatically major.

Major intervals generally have a bright, “happy” sound, which is part of why the major scale itself is often described as sounding cheerful or uplifting.

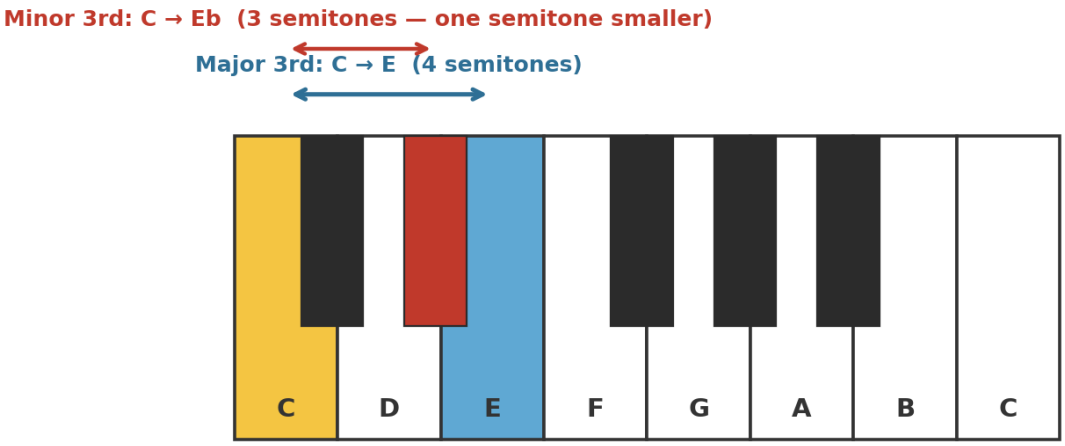


Figure 4.1 — C to E is a major 3rd (4 semitones) — the note E appears naturally in the C major scale.

EXAMPLE

Example	Interval Name	How We Count It
C to D	Major 2nd	2 semitones
C to E	Major 3rd	4 semitones

Example	Interval Name	How We Count It
C to A	Major 6th	9 semitones
C to B	Major 7th	11 semitones

PRACTICAL EXERCISE

1. Write out the C major scale, then identify which note is a major 3rd above C.
2. Play a major 2nd starting on G. Which note did you land on?
3. Play a major 6th starting on F. Which note did you land on?
4. In your own words, explain how the major scale helps you find major intervals quickly.

4.2 Minor

DEFINITION

A minor interval is one semitone smaller than a major interval of the same number, and only applies to 2nds, 3rds, 6ths, and 7ths.

EXPLANATION

The word “minor” means “smaller”. To turn any major interval into a minor interval, simply lower the top note by one semitone — the interval number stays exactly the same, but the sound becomes slightly smaller and, to many listeners, slightly sadder or more emotional.

For example, a major 3rd (C to E) becomes a minor 3rd (C to Eb) once we lower the E by one semitone. The letter-name count is still 3 (C, D, E/Eb), so the number stays a 3rd — only the exact size in semitones has changed.

Minor intervals are essential to minor scales and minor chords, which is why so much music described as sounding “sad”, “mysterious”, or “serious” relies heavily on minor 3rds and minor 6ths.

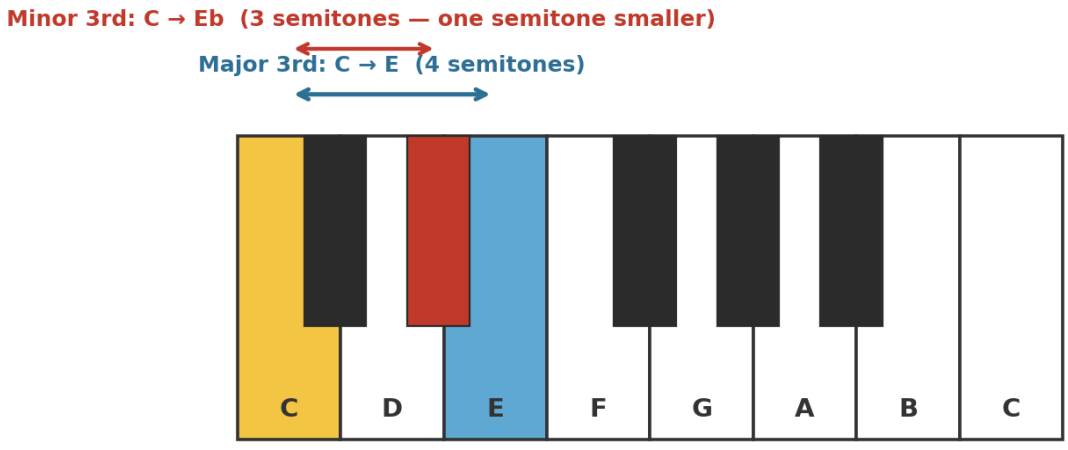


Figure 4.2 — C to Eb is a minor 3rd (3 semitones) — exactly one semitone smaller than the major 3rd, C to E.

EXAMPLE

Example	Interval Name	How We Count It
C to Db	Minor 2nd	1 semitone
C to Eb	Minor 3rd	3 semitones
C to Ab	Minor 6th	8 semitones
C to Bb	Minor 7th	10 semitones

PRACTICAL EXERCISE

1. Take the major 3rd from C (C to E) and lower it by one semitone. What minor interval do you now have?
2. Play a minor 2nd starting on D. Which note did you land on?
3. Play a minor 7th starting on G. Which note did you land on?
4. Explain, in your own words, the difference in sound between a major 3rd and a minor 3rd.

4.3 Perfect

DEFINITION

A perfect interval is an especially stable, consonant interval that applies only to unisons, 4ths, 5ths, and octaves — these intervals do not have major or minor versions.

EXPLANATION

Unlike 2nds, 3rds, 6ths, and 7ths, the unison, 4th, 5th, and octave behave differently — they are never called “major” or “minor”. Instead, their natural, most stable size is simply called “perfect”.

The word “perfect” here does not mean “flawless” in an everyday sense — it is a historical music term describing intervals that sound especially smooth, stable, and consonant (pleasant when played together), because of the simple mathematical relationship between their sound frequencies.

A perfect unison is 0 semitones (the same note), a perfect 4th is 5 semitones, a perfect 5th is 7 semitones, and a perfect octave is 12 semitones. These four intervals form the stable “skeleton” that much of Western music is built around.

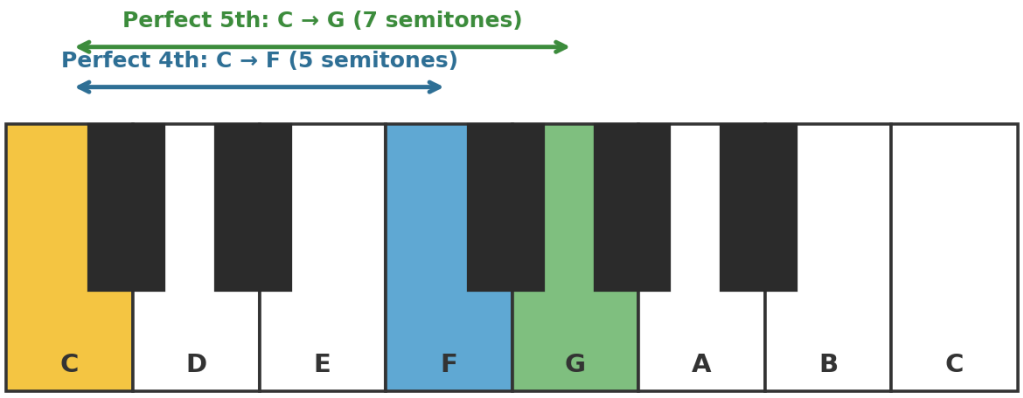


Figure 4.3 — The perfect 4th (C to F, 5 semitones) and perfect 5th (C to G, 7 semitones) are both stable, "open"-sounding intervals.

EXAMPLE

Example	Interval Name	How We Count It
C to C	Perfect Unison	0 semitones
C to F	Perfect 4th	5 semitones
C to G	Perfect 5th	7 semitones
C to C (next)	Perfect Octave	12 semitones

PRACTICAL EXERCISE

1. List the four interval numbers that use the word "perfect" instead of "major" or "minor".
2. Play a perfect 5th starting on D. Which note did you land on?
3. Play a perfect 4th starting on A. Which note did you land on?
4. Why do you think perfect intervals are described as sounding "stable" compared to major or minor intervals?

4.4 Augmented

DEFINITION

An augmented interval is any major or perfect interval that has been made one semitone larger.

EXPLANATION

The word "augmented" means "made bigger". To create an augmented interval, take a major interval (for 2nds, 3rds, 6ths, 7ths) or a perfect interval (for unisons, 4ths, 5ths, octaves) and raise the top note by one semitone, while keeping the same letter-name distance.

For example, a perfect 5th (C to G, 7 semitones) becomes an augmented 5th (C to G#, 8 semitones) once we raise G by one semitone to G#. Note that the letter names are still C and G — only G has been sharpened — so the interval number stays a 5th, but the quality changes to augmented.

Augmented intervals often create a feeling of tension or an unexpected "stretch" in the sound, and are used by composers to add colour, surprise, or drama, especially in jazz and 20th-century classical music.

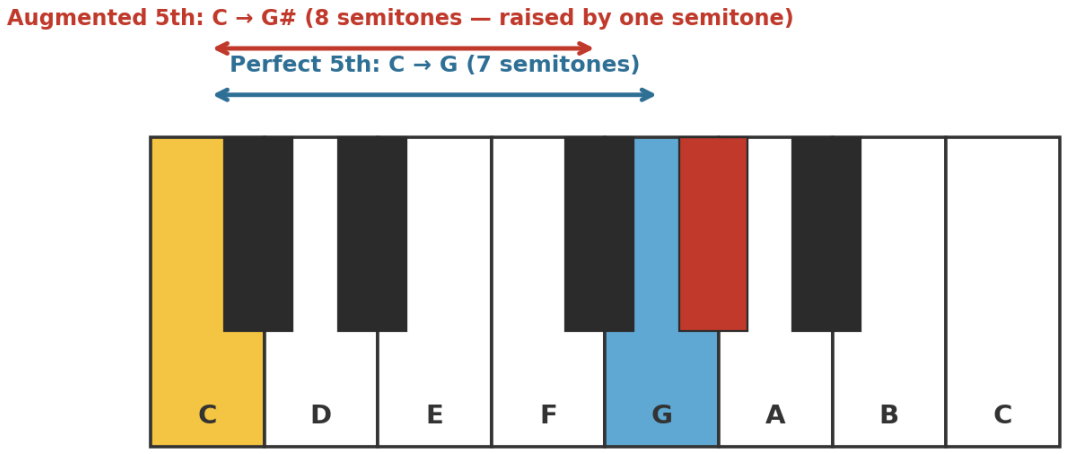


Figure 4.4 — C to G# is an augmented 5th (8 semitones) — one semitone larger than the perfect 5th, C to G.

EXAMPLE

Example	Interval Name	How We Count It
C to G#	Augmented 5th	8 semitones (perfect 5th + 1)
C to F#	Augmented 4th	6 semitones (perfect 4th + 1)
C to E#	Augmented 3rd	5 semitones (major 3rd + 1)

PRACTICAL EXERCISE

1. Start with a perfect 4th from C to F. Raise the top note by one semitone. What is this new interval called?
2. Start with a major 2nd from C to D. Raise the top note by one semitone. What is this new interval called?
3. Play C to G, then play C to G#. Describe the difference in how each one sounds.
4. Explain, in your own words, why an augmented interval is always one semitone larger than a major or perfect interval of the same number.

4.5 Diminished

DEFINITION

A diminished interval is any minor or perfect interval that has been made one semitone smaller.

EXPLANATION

The word “diminished” means “made smaller”. To create a diminished interval, take a minor interval (for 2nds, 3rds, 6ths, 7ths) or a perfect interval (for unisons, 4ths, 5ths, octaves) and lower the top note by one semitone, while keeping the same letter-name distance.

For example, a perfect 5th (C to G, 7 semitones) becomes a diminished 5th (C to Gb, 6 semitones) once we lower G by one semitone to Gb. The letter names are still C and G — only G has been flattened — so the interval number stays a 5th, but the quality changes to diminished.

The diminished 5th is especially famous — it is also called the “tritone”, and has a very tense, unstable sound that composers have used for centuries to create suspense. It is sometimes nicknamed “the devil's interval” because of how uneasy it can sound.

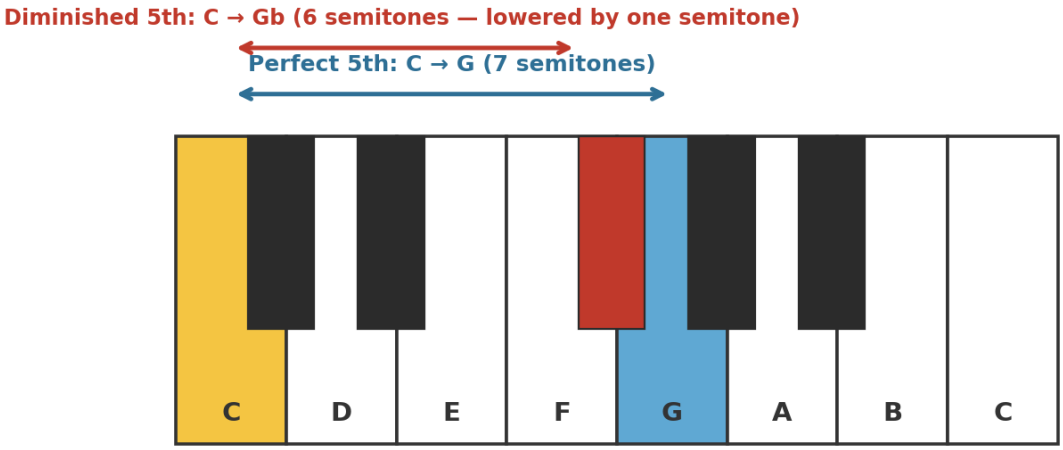


Figure 4.5 — C to Gb is a diminished 5th (6 semitones) — one semitone smaller than the perfect 5th, C to G.

EXAMPLE

Example	Interval Name	How We Count It
C to Gb	Diminished 5th	6 semitones (perfect 5th − 1)
C to Fb	Diminished 4th	4 semitones (perfect 4th − 1)
C to Ebb	Diminished 3rd	2 semitones (minor 3rd − 1)

PRACTICAL EXERCISE

1. Start with a perfect 5th from C to G. Lower the top note by one semitone. What is this new interval called?
2. Start with a minor 7th from C to Bb. Lower the top note by one semitone. What is this new interval called?
3. Play C to G, then play C to Gb. Describe the difference in how each one sounds.
4. Look up why the diminished 5th is nicknamed “the devil's interval”, and write one sentence about what you found.

Chapter 5

Major & Minor Intervals

In Chapter 4, we met the idea of major and minor as interval qualities. Now we look closely at each specific major and minor interval — 2nds, 3rds, 6ths, and 7ths — one at a time, so you become confident recognising, playing, and hearing every one of them.

5.1 Major 2nd

DEFINITION

A major 2nd spans two semitones between two notes that are one letter name apart.

EXPLANATION

A major 2nd is simply a whole tone — the term you already learned in Chapter 2. From C to D, we move up two semitones (C → C# → D), passing through one note along the way.

Major 2nds are everywhere in music — most steps of a major scale are major 2nds. This interval has a bright, simple, "stepping" sound, and it is usually the very first interval beginner musicians learn to sing or play accurately.

You can find a major 2nd by starting on any note and moving to the note two keys away, skipping exactly one key (black or white) in between.

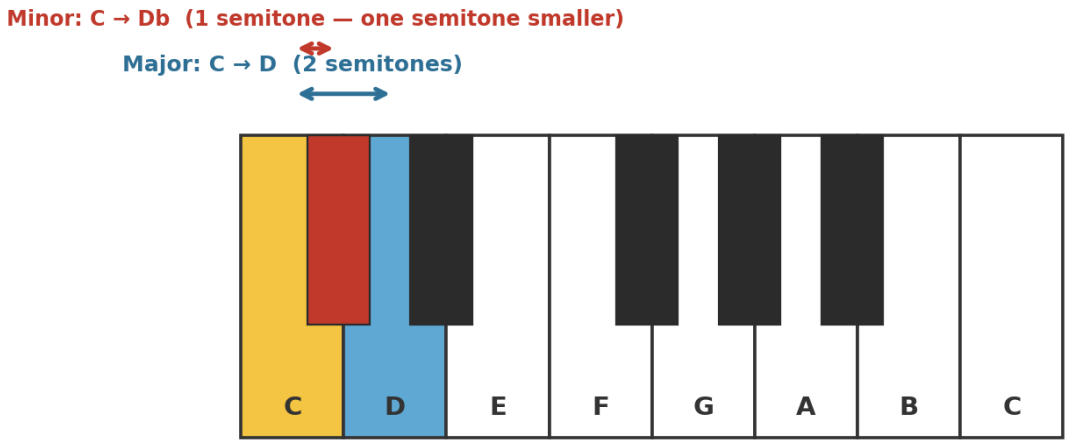


Figure 5.1 — C to D is a major 2nd (2 semitones), shown alongside the minor 2nd (C to Db) for comparison.

EXAMPLE

Example	Interval Name	How We Count It
C to D	Major 2nd	2 semitones
F to G	Major 2nd	2 semitones
G to A	Major 2nd	2 semitones

PRACTICAL EXERCISE

1. Play a major 2nd starting on D. Which note did you land on?
2. Sing the first two notes of “Happy Birthday” (“Hap-py”). Is this interval a major 2nd?
3. Name three major 2nds you can find within the C major scale.
4. Explain, using what you learned in Chapter 2, why a major 2nd is the same thing as a whole tone.

5.2 Minor 2nd

DEFINITION

A minor 2nd spans one semitone between two notes that are one letter name apart.

EXPLANATION

A minor 2nd is simply a semitone — the smallest interval in Western music, which you first learned about in Chapter 2. From C to Db, we move up just one semitone, with no note in between at all.

The minor 2nd has a very tight, tense, “rubbing” sound, because the two notes are so close together that they almost blend into one another. This interval is often used to create suspense, unease, or drama — the famous shower-scene music from the film “Psycho” is built almost entirely from minor 2nds.

You can find a minor 2nd by starting on any note and moving to the very next key beside it — no keys skipped at all.

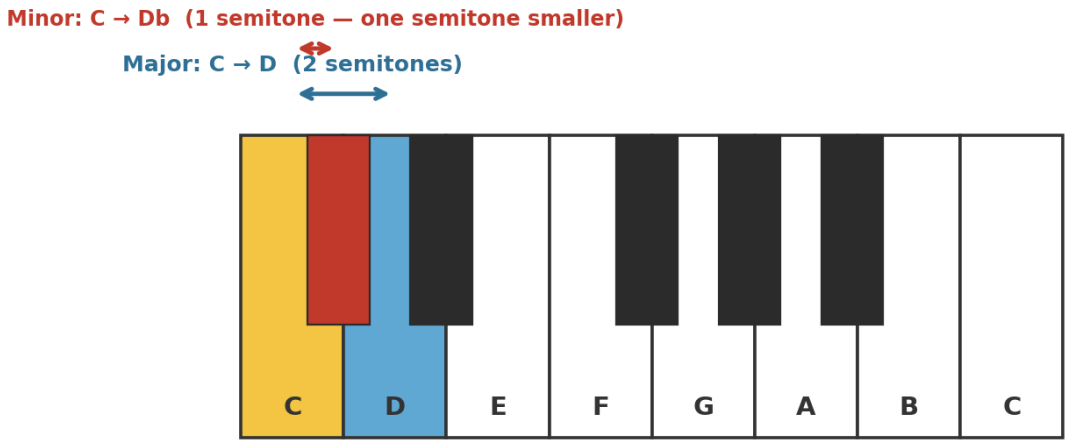


Figure 5.2 — C to Db is a minor 2nd (1 semitone) — the smallest interval possible, one semitone smaller than the major 2nd.

EXAMPLE

Example	Interval Name	How We Count It
C to Db	Minor 2nd	1 semitone
E to F	Minor 2nd	1 semitone

Example	Interval Name	How We Count It
B to C	Minor 2nd	1 semitone

PRACTICAL EXERCISE

1. Play a minor 2nd starting on G. Which note did you land on?
2. Name the two places within the C major scale where the interval between neighbouring notes is naturally a minor 2nd (no sharps or flats needed).
3. Play a major 2nd, then a minor 2nd, right after each other. Describe the difference in tension you hear.
4. Why do you think film composers might use minor 2nds to create a feeling of suspense?

5.3 Major 3rd

DEFINITION

A major 3rd spans four semitones between two notes that are two letter names apart.

EXPLANATION

A major 3rd is one of the most important intervals in all of music. From C to E, we move up four semitones, skipping over the note D along the way.

The major 3rd is the interval that gives major chords their bright, happy sound — every basic major chord contains a major 3rd between its root and middle note.

Musicians often train their ear to recognise the major 3rd first, because it sounds noticeably different from the minor 3rd and appears constantly in melodies, chords, and basslines across almost every style of music.

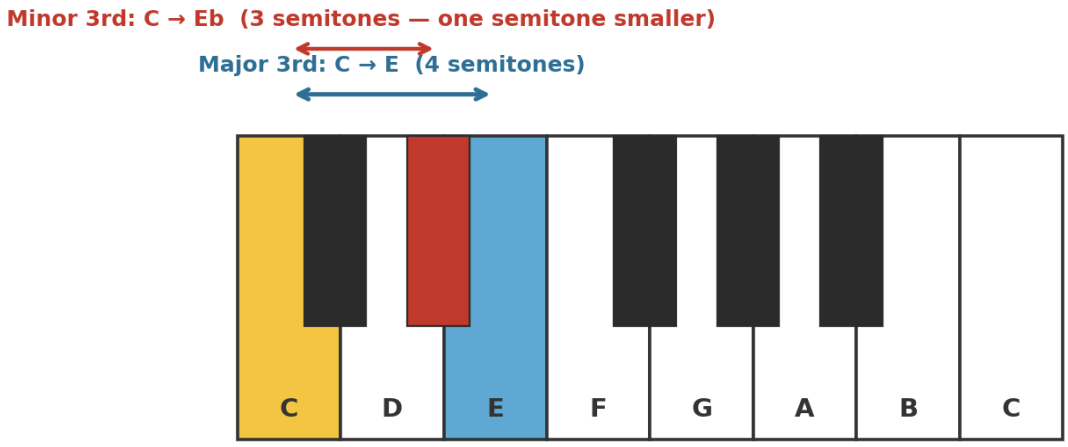


Figure 5.3 — C to E is a major 3rd (4 semitones), shown alongside the minor 3rd (C to Eb) for comparison.

EXAMPLE

Example	Interval Name	How We Count It
C to E	Major 3rd	4 semitones
F to A	Major 3rd	4 semitones
G to B	Major 3rd	4 semitones

PRACTICAL EXERCISE

1. Play a major 3rd starting on D. Which note did you land on?
2. Play a C major chord (C, E, G). Can you identify the major 3rd hiding inside it?
3. Sing the opening two notes of “When the Saints Go Marching In”. Is this interval a major 3rd?
4. Name three major 3rds you can find by starting on different white keys of the C major scale.

5.4 Minor 3rd

DEFINITION

A minor 3rd spans three semitones between two notes that are two letter names apart.

EXPLANATION

A minor 3rd is one semitone smaller than a major 3rd. From C to Eb, we move up three semitones, skipping over the note D along the way, just like the major 3rd — but landing one semitone lower.

The minor 3rd is the interval that gives minor chords their darker, more emotional sound — every basic minor chord contains a minor 3rd between its root and middle note.

Many musicians describe the minor 3rd as sounding “sadder” or “more serious” than the major 3rd, even though both intervals span the same letter-name distance — this shows just how much a single semitone can change the emotional colour of music.

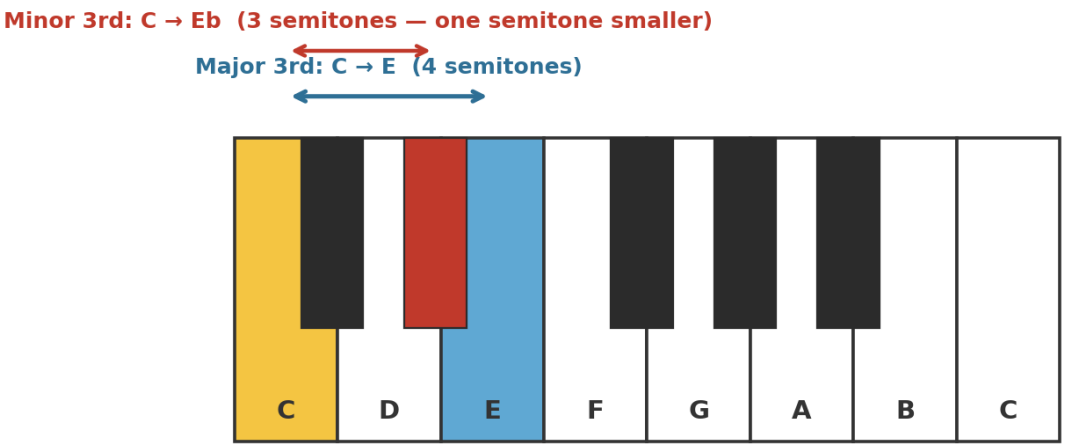


Figure 5.4 — C to Eb is a minor 3rd (3 semitones) — one semitone smaller than the major 3rd, C to E.

EXAMPLE

Example	Interval Name	How We Count It
C to Eb	Minor 3rd	3 semitones
D to F	Minor 3rd	3 semitones
A to C	Minor 3rd	3 semitones

PRACTICAL EXERCISE

1. Play a minor 3rd starting on G. Which note did you land on?
2. Play a C minor chord (C, Eb, G). Can you identify the minor 3rd hiding inside it?
3. Play a major 3rd and a minor 3rd back to back, starting on the same note. Describe the emotional difference you hear.
4. Find one minor 3rd that occurs naturally within the C major scale, without adding any sharps or flats.

5.5 Major 6th

DEFINITION

A major 6th spans nine semitones between two notes that are five letter names apart.

EXPLANATION

A major 6th is a wide, warm-sounding interval. From C to A, we move up nine semitones, counting through five letter names in between (C, D, E, F, G, A).

One helpful shortcut: a major 6th is the exact inversion of a minor 3rd — if you take a minor 3rd and flip it (moving the bottom note up an octave instead), you get a major 6th. We will explore inversion more closely in a later chapter.

The major 6th has a bright, spacious sound and is very commonly used in harmony singing, where a second singer adds a note a 6th above or below the main melody to create a rich, pleasant blend.

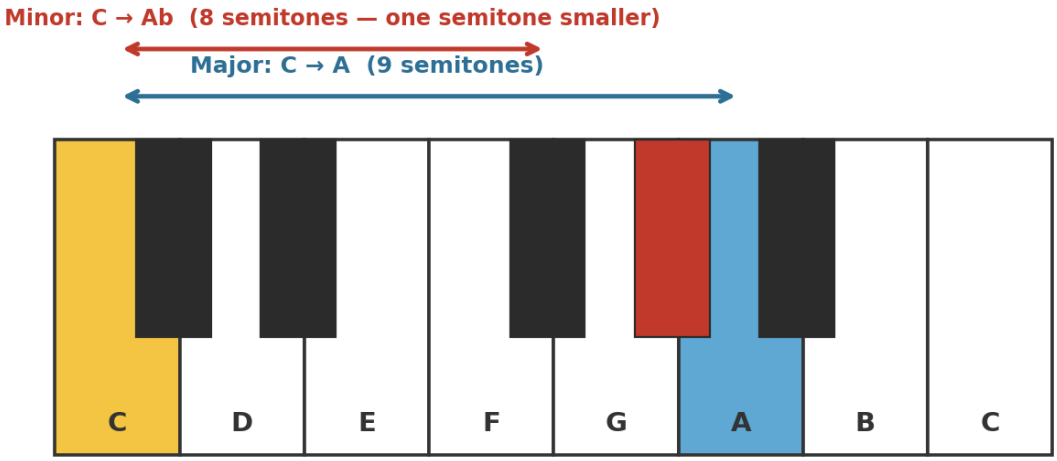


Figure 5.5 — C to A is a major 6th (9 semitones), shown alongside the minor 6th (C to Ab) for comparison.

EXAMPLE

Example	Interval Name	How We Count It
C to A	Major 6th	9 semitones
F to D	Major 6th	9 semitones
G to E	Major 6th	9 semitones

PRACTICAL EXERCISE

1. Play a major 6th starting on D. Which note did you land on?
2. Sing the opening two notes of “My Bonnie Lies Over the Ocean” (“My Bon-”). Is this interval a major 6th?
3. If you know another singer or musician, try singing a simple melody together a 6th apart. How does it sound?
4. Explain, in your own words, what it means for a major 6th to be the “inversion” of a minor 3rd.

5.6 Minor 6th

DEFINITION

A minor 6th spans eight semitones between two notes that are five letter names apart.

EXPLANATION

A minor 6th is one semitone smaller than a major 6th. From C to Ab, we move up eight semitones, still counting through five letter names (C, D, E, F, G, A/Ab).

The minor 6th has a more emotional, sometimes yearning or bittersweet sound compared to the brighter major 6th. It is often used in film scores and ballads to add a touch of longing or nostalgia.

Just as the major 6th is the inversion of the minor 3rd, the minor 6th is the inversion of the major 3rd — these paired relationships will make more sense once you study interval inversion in a later chapter.

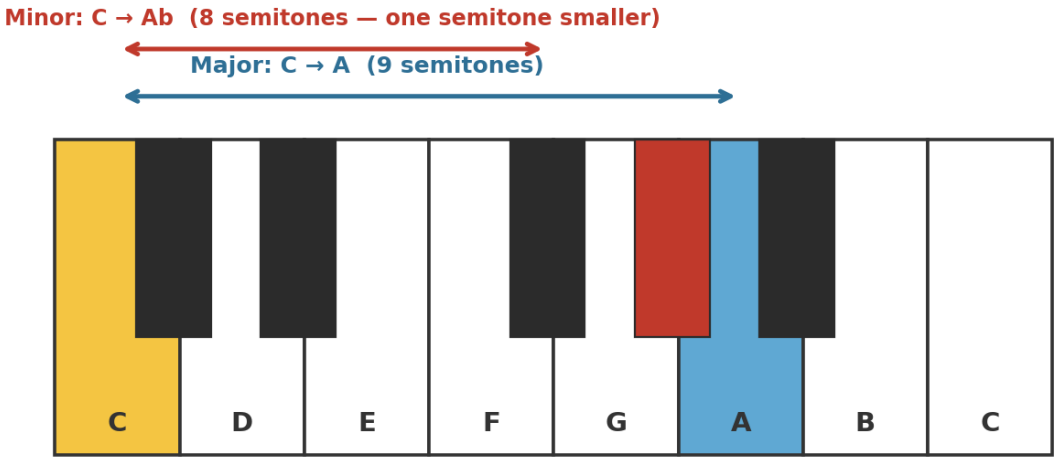


Figure 5.6 — C to Ab is a minor 6th (8 semitones) — one semitone smaller than the major 6th, C to A.

EXAMPLE

Example	Interval Name	How We Count It
C to Ab	Minor 6th	8 semitones
E to C	Minor 6th	8 semitones
A to F	Minor 6th	8 semitones

PRACTICAL EXERCISE

1. Play a minor 6th starting on G. Which note did you land on?
2. Play a major 6th and a minor 6th back to back, starting on the same note. Describe the difference in feeling you hear.
3. Listen to a slow, emotional song you know. Try to spot a moment where the melody leaps by what might be a 6th.
4. Explain why film composers might reach for a minor 6th when they want to create a feeling of longing.

5.7 Major 7th

DEFINITION

A major 7th spans eleven semitones between two notes that are six letter names apart — just one semitone short of a full octave.

EXPLANATION

A major 7th is a wide, tense interval. From C to B, we move up eleven semitones, counting through six letter names in between (C, D, E, F, G, A, B).

Because it sits just one semitone below the octave, the major 7th creates a strong feeling of "almost there" — the ear expects it to resolve upward to the octave, which makes it feel bright but slightly unsettled at the same time.

The major 7th is a favourite interval in jazz harmony, often added to chords to create a lush, sophisticated "jazzy" sound (these are called "major 7th chords").

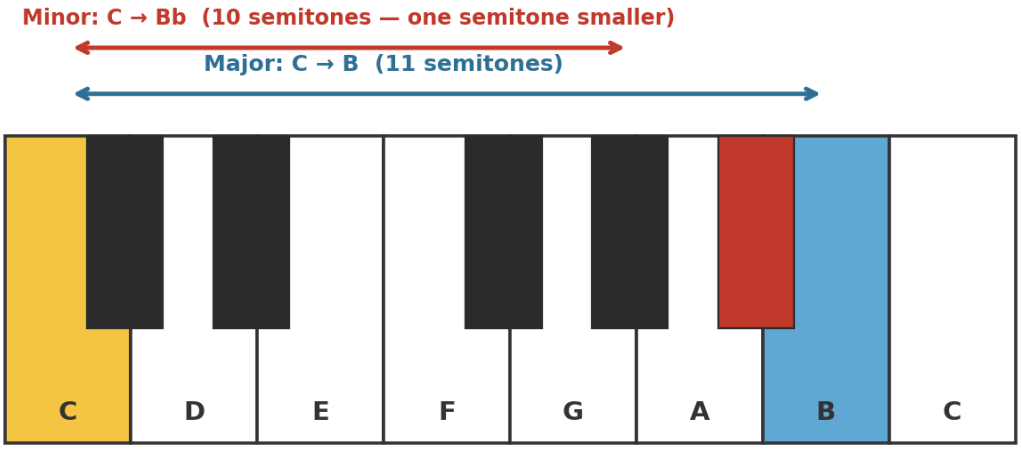


Figure 5.7 — C to B is a major 7th (11 semitones), shown alongside the minor 7th (C to Bb) for comparison.

EXAMPLE

Example	Interval Name	How We Count It
C to B	Major 7th	11 semitones
F to E	Major 7th	11 semitones
G to F#	Major 7th	11 semitones

PRACTICAL EXERCISE

1. Play a major 7th starting on D. Which note did you land on?
2. Play C to B, then resolve upward to C (the octave). Can you feel the tension release?
3. Listen to a jazz recording and see if you can spot the lush, dreamy sound of a major 7th chord.
4. Explain why the major 7th might feel like it "wants" to move up to the octave.

5.8 Minor 7th

DEFINITION

A minor 7th spans ten semitones between two notes that are six letter names apart.

EXPLANATION

A minor 7th is one semitone smaller than a major 7th. From C to Bb, we move up ten semitones, still counting through six letter names (C, D, E, F, G, A, B/Bb).

The minor 7th has a bluesy, soulful sound and is extremely common in blues, jazz, funk, and pop music. It is the interval that gives “dominant 7th” and “minor 7th” chords their distinctive, slightly relaxed tension.

Unlike the major 7th, which feels like it urgently wants to resolve to the octave, the minor 7th feels more relaxed and settled, which is part of why it is used so comfortably throughout entire songs rather than just as a passing moment of tension.

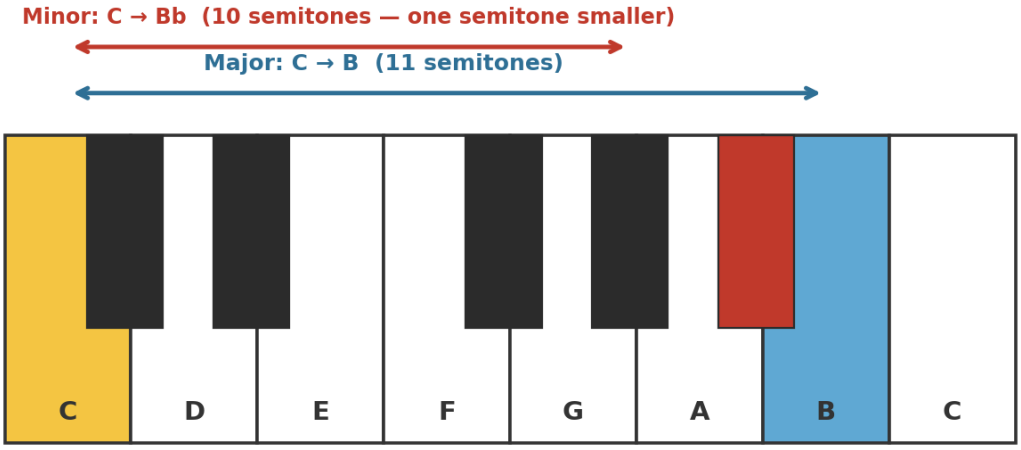


Figure 5.8 — C to Bb is a minor 7th (10 semitones) — one semitone smaller than the major 7th, C to B.

EXAMPLE

Example	Interval Name	How We Count It
C to Bb	Minor 7th	10 semitones
D to C	Minor 7th	10 semitones
G to F	Minor 7th	10 semitones

PRACTICAL EXERCISE

1. Play a minor 7th starting on E. Which note did you land on?
2. Play a major 7th and a minor 7th back to back, starting on the same note. Which one sounds more relaxed to you?
3. Listen to a blues or funk song and see if you can hear the soulful sound of a minor 7th (or a 7th chord).
4. Summarise, in your own words, the difference in feeling between a major 7th and a minor 7th.